Wireshark Analysis Screenshots

seeded_traffic.pcap — protocol filtering, packet inspection, and OSINT lookups

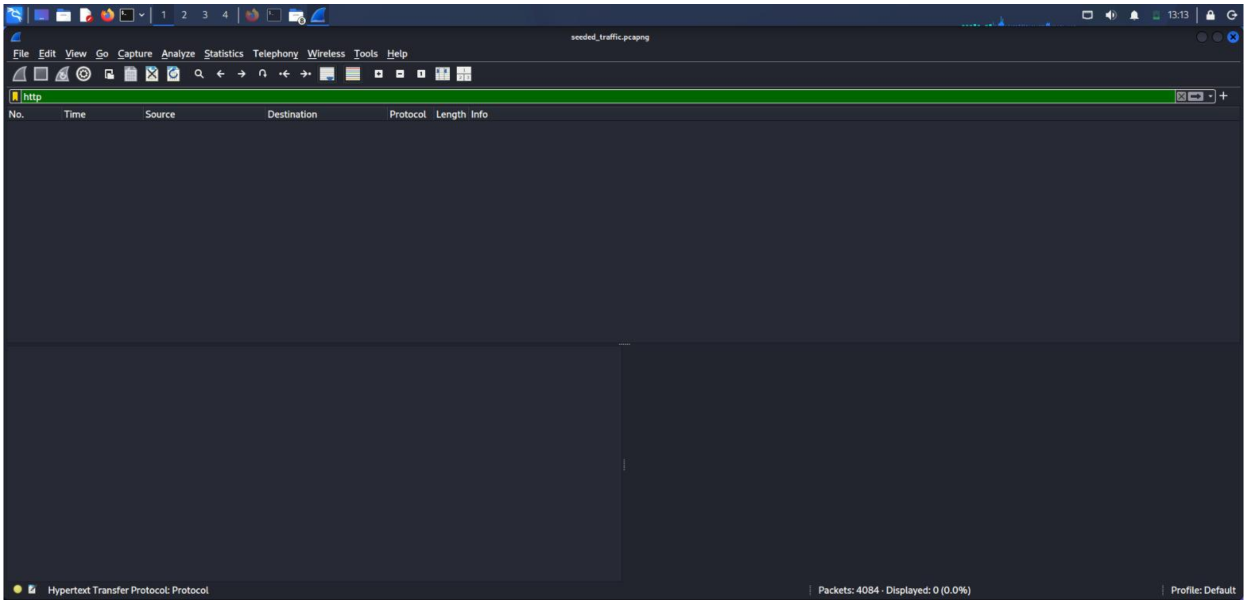
Overview

This document is a supporting evidence log for the Network Traffic Analysis & Threat Detection project. It captures each Wireshark filter step used to inspect seeded_traffic.pcap, alongside the two external OSINT lookups (VirusTotal and Whois) performed on the domain and IP address flagged during analysis.

Screenshots are grouped in the order the investigation was carried out: first isolating traffic by protocol (HTTP, DNS, FTP, TCP), then confirming the plaintext FTP credentials at the byte level, then identifying the port-scan signature, and finally cross-referencing the flagged domain and IP against public threat-intelligence sources.

Contents

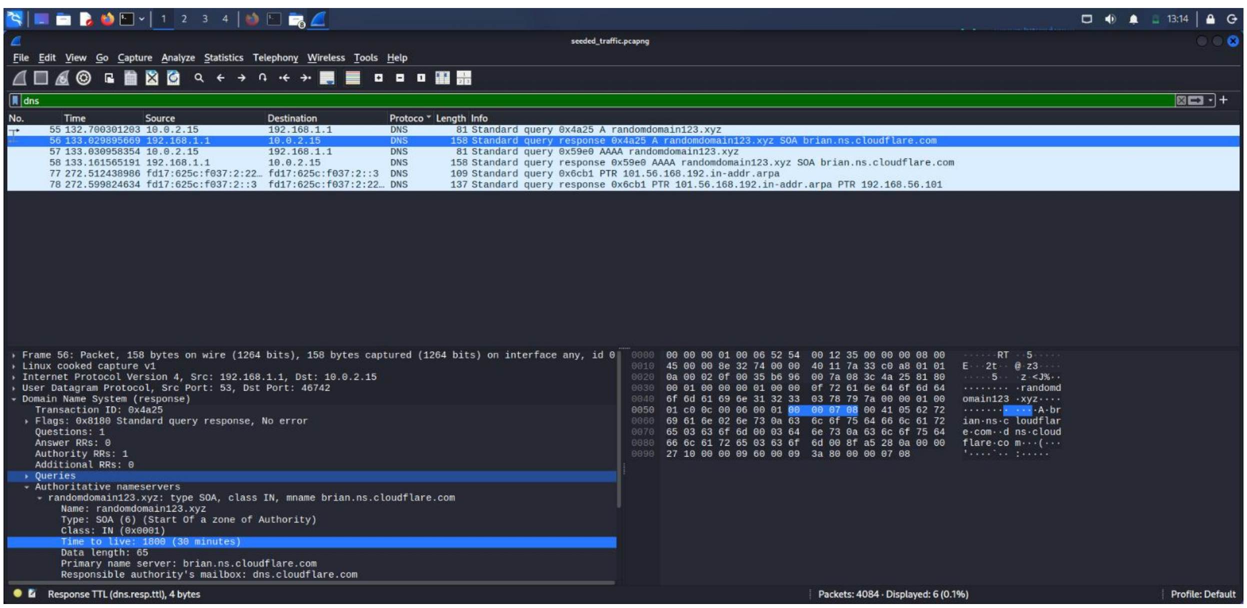
1. HTTP filter



Filtering by "http" returns 0 packets — no web browsing traffic present in this capture.

Technical detail: Status bar confirms "Displayed: 0 (0.0%)" out of 4,084 total packets — the absence is a clean negative result, not a missed filter.

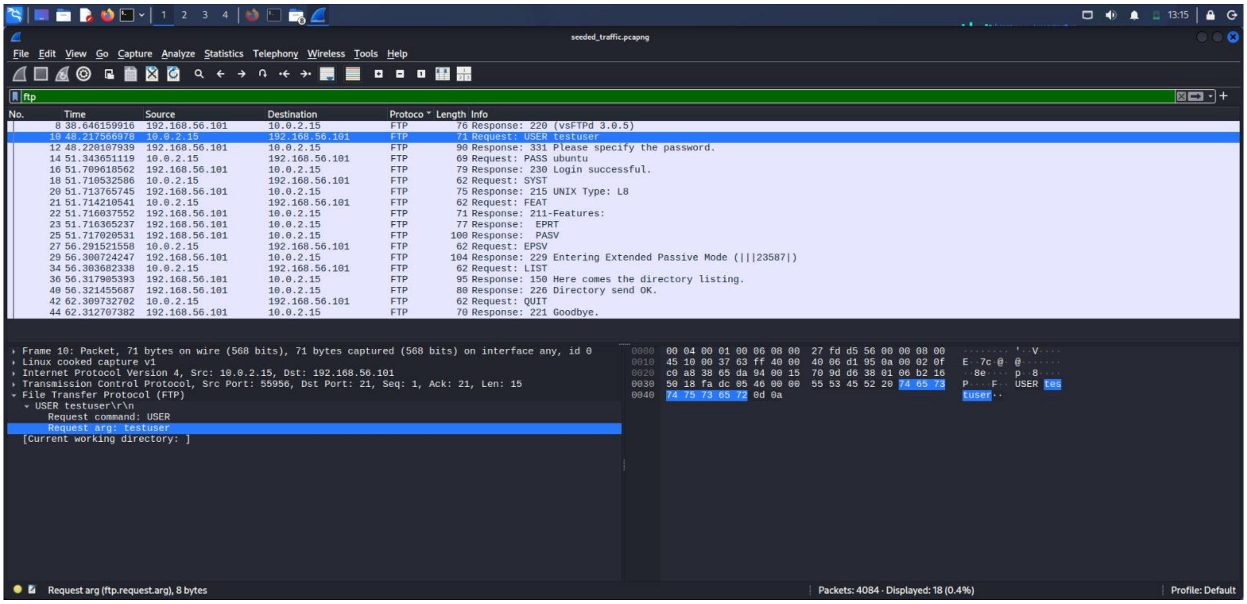
2. DNS filter



Filtering by "dns" shows the query to randomdomain123.xyz, plus a reverse-lookup entry. The Authoritative nameservers section confirms no valid A record exists for the domain.

Technical detail: Transaction ID 0x4a25, Flags 0x8180 (standard query response, no error), Answer RRs: 0 — the server responded but returned zero answer records, only an SOA record naming brian.ns.cloudflare.com as authority.

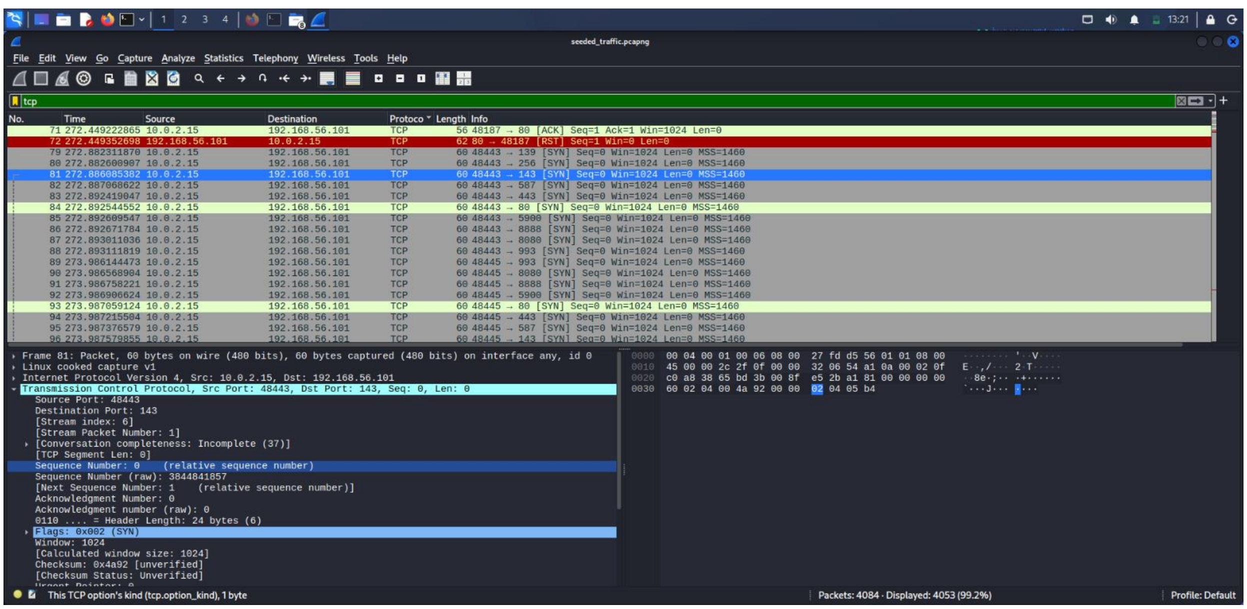
3. FTP filter



Filtering by "ftp" reveals the full login session: USER testuser, PASS ubuntu, login success, directory listing, and session close.

Technical detail: 18 packets total in the session (Displayed: 18, 0.4%), spanning response code 220 (service ready) through to 221 (goodbye) — a complete, unencrypted control-channel conversation.

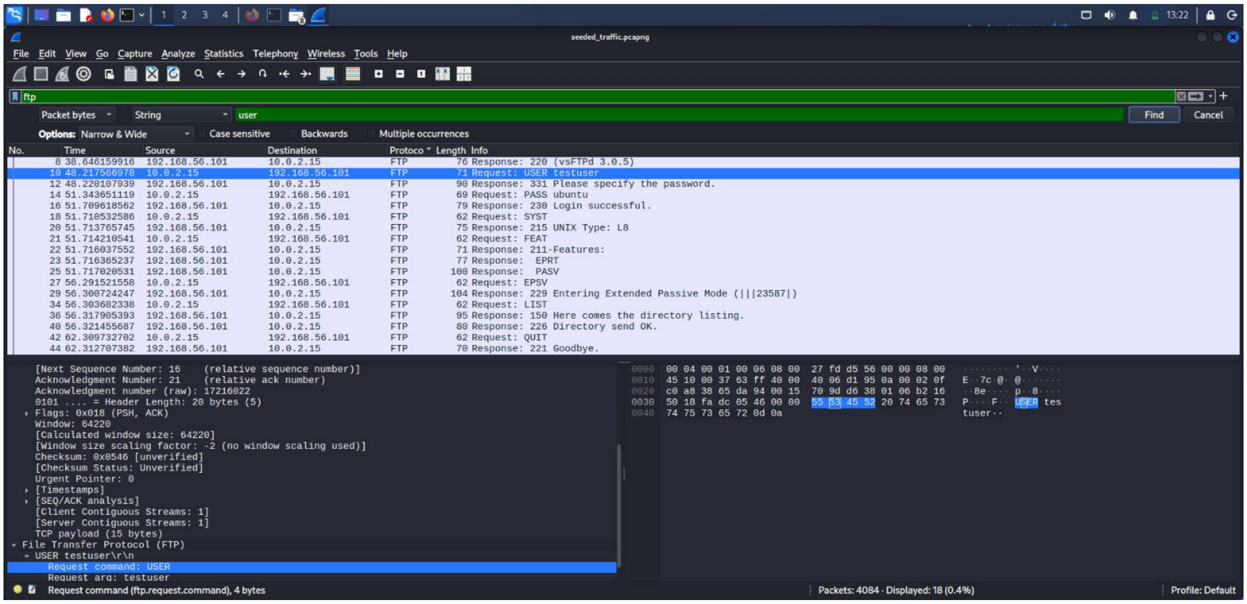
4. TCP filter



Filtering by "tcp" shows the raw volume of traffic, including the rapid sequence of SYN packets to many destination ports generated by the Nmap scan.

Technical detail: Displayed: 4,053 packets (99.2% of the capture) — confirms TCP is by far the dominant protocol in this file, consistent with the scan being the bulk of the seeded activity.

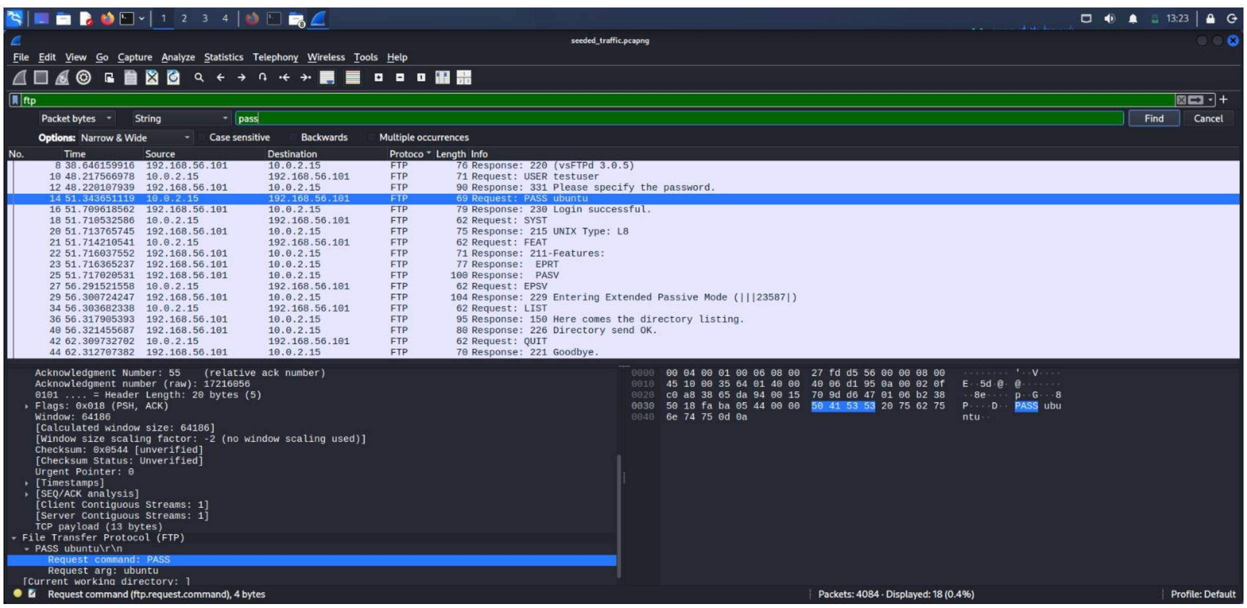
5. FTP packet bytes — "user" search



Searching packet bytes for "user" highlights the USER command and confirms the username "testuser" is stored in plaintext in the packet payload.

Technical detail: Hex offset 0x0030–0x0040 shows the ASCII bytes "USER tes...tuser" directly in the payload — no encoding or obfuscation of any kind.

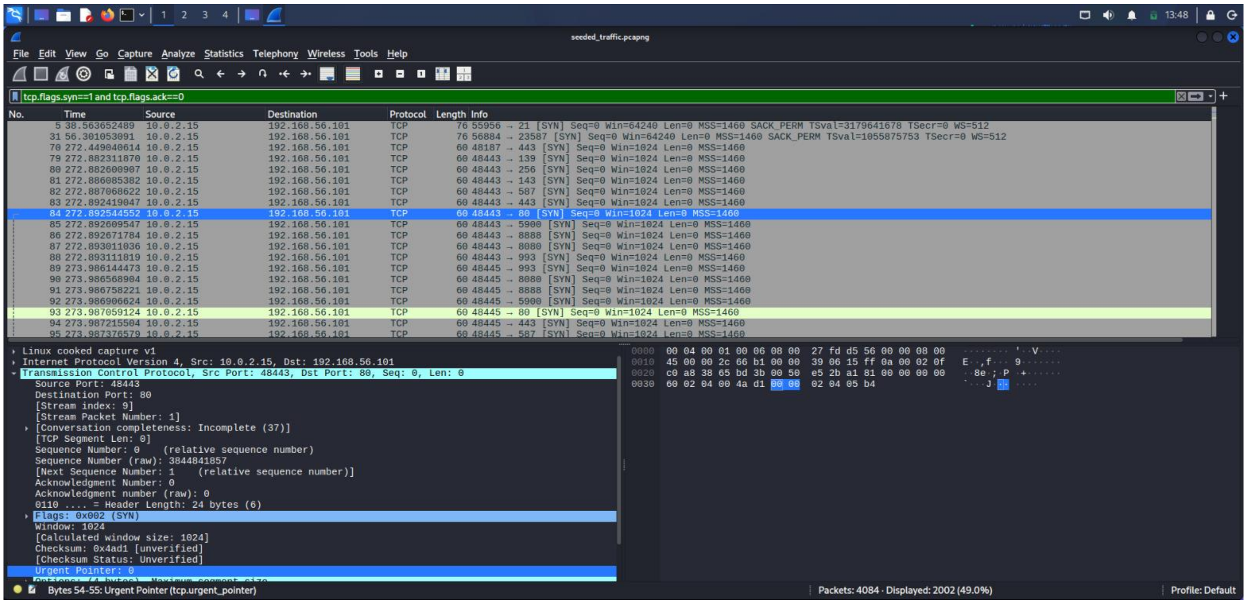
6. FTP packet bytes — "pass" search



Searching packet bytes for "pass" highlights the PASS command and confirms the password "ubuntu" is stored in plaintext in the packet payload.

Technical detail: Request command: PASS, Request arg: ubuntu — visible directly in Wireshark's dissected FTP layer, not just the hex dump, confirming Wireshark itself parses the credential without any decryption step.

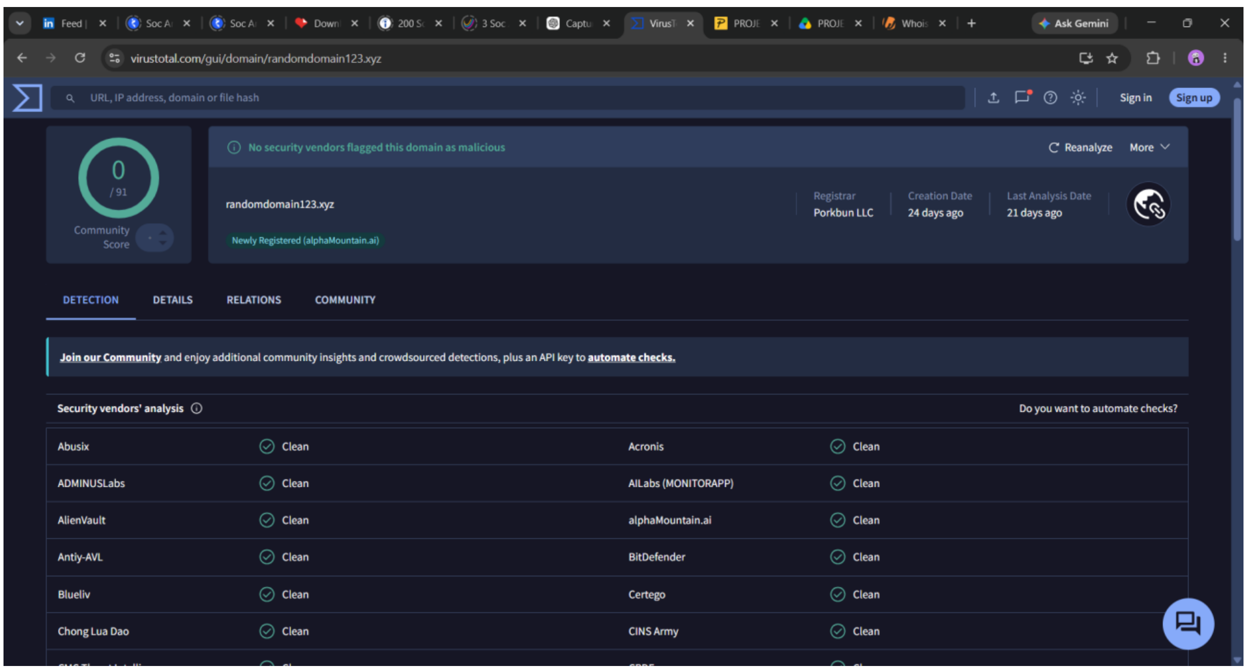
7. SYN scan filter



Filtering by tcp.flags.syn==1 and tcp.flags.ack==0 isolates 2,002 SYN packets sent to 192.168.56.101 across many destination ports within about one second — the signature of an Nmap -sS scan.

Technical detail: Displayed: 2,002 (49.0% of the entire capture) — nearly half of all packets in the file are scan-related SYN probes, each with Win=1024 and identical MSS=1460, a consistent fingerprint of a single automated scanning tool.

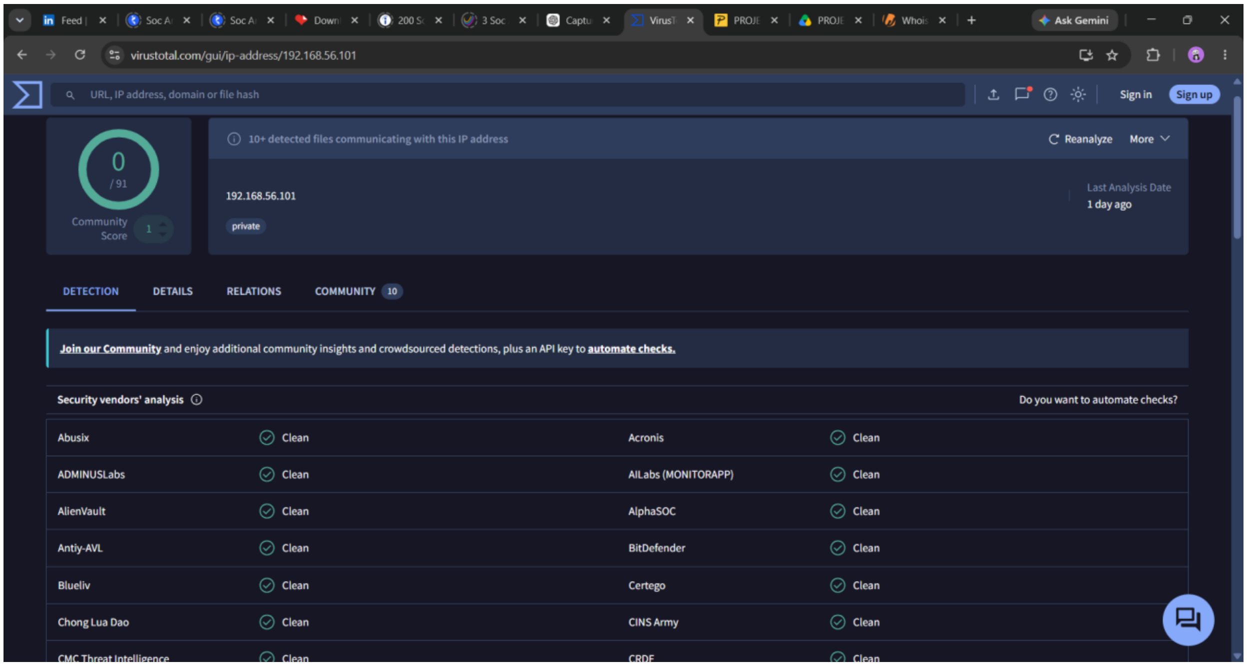
8. VirusTotal — randomdomain123.xyz



VirusTotal shows 0/91 vendors flag this domain as malicious, but tags it "Newly Registered" (24 days old, registrar Porkbun LLC) — a mild indicator worth monitoring despite the clean detection result.

Technical detail: 91 separate security vendors were queried and returned zero detections — a genuinely clean scan, but one performed against a domain still within its first month of existence, when detection engines have the least historical data to work from.

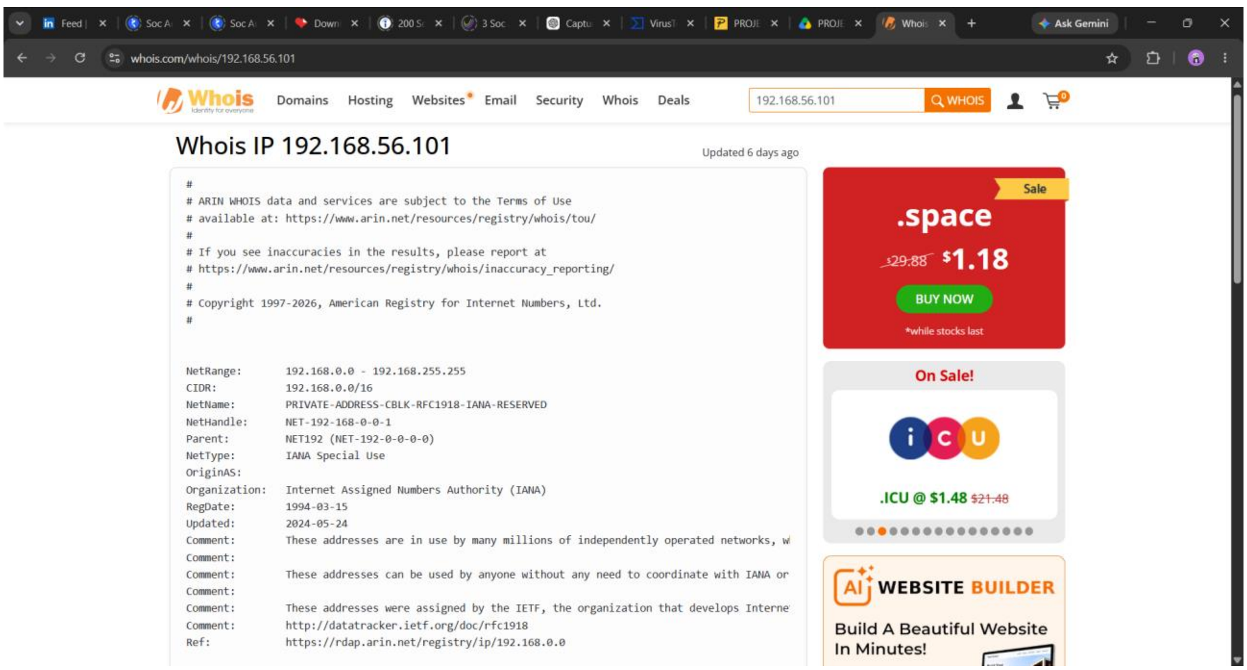
9. VirusTotal — 192.168.56.101



VirusTotal shows 0/91 detections and tags the address "private." It notes "10+ detected files communicating with this IP" historically — expected, since this private range is reused across countless unrelated networks worldwide and is not evidence of compromise in this lab.

Technical detail: Community Score of 1 and a "private" tag confirm VirusTotal itself recognizes this as a non-routable address; the 10+ file association reflects its global dataset, not any traffic generated in this lab.

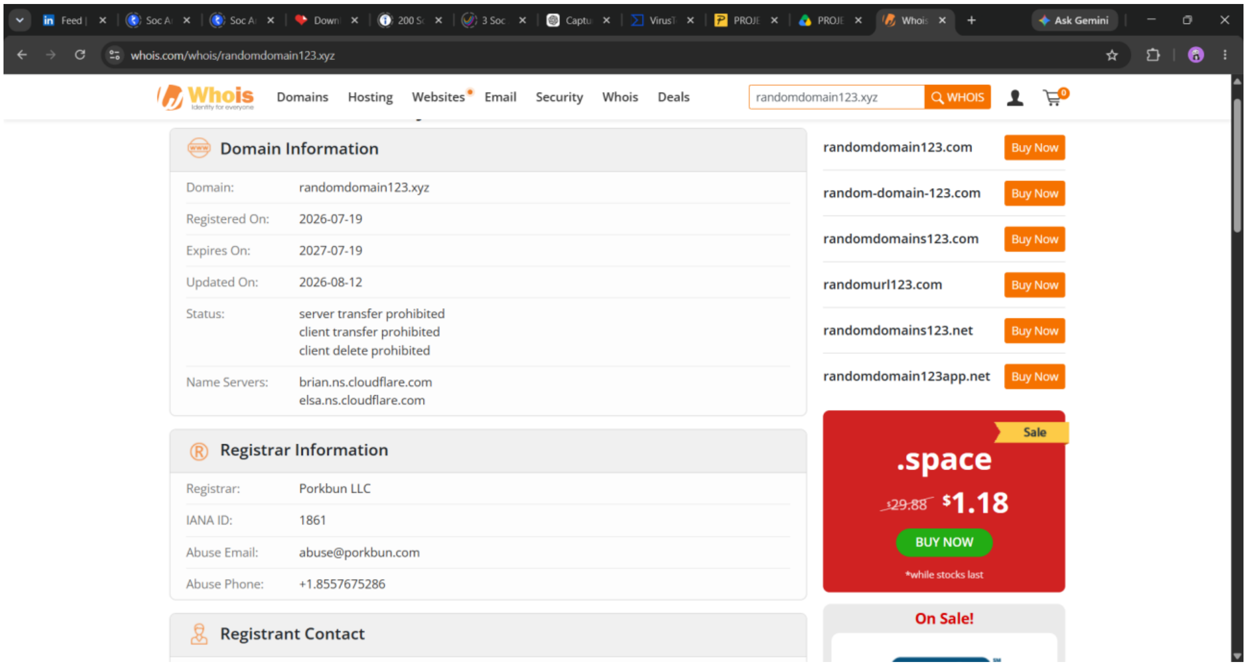
10. Whois — randomdomain123.xyz



Whois confirms the domain is actively registered: registered 2026-07-19 through Porkbun LLC, using Cloudflare nameservers, with transfer/delete protection enabled.

Technical detail: Registered 2026-07-19, expires 2027-07-19 — a standard one-year registration term, with server/client transfer and delete locks enabled, typical of a legitimately (if recently) registered domain rather than a disposable throwaway.

11. Whois — 192.168.56.101



Whois confirms 192.168.56.101 falls within the IANA-reserved private address block (192.168.0.0/16), used by millions of independently operated private networks worldwide — consistent with it being an internal lab address, not a publicly registered host.

Technical detail: NetType: IANA Special Use, RegDate: 1994-03-15 — the entire /16 block was reserved by the IETF (RFC 1918) three decades ago specifically so it would never need individual per-address registration.

Summary

Across these eleven screenshots, three protocol-level findings and two OSINT lookups were documented: FTP transmits login credentials in plaintext (testuser / ubuntu), a DNS query was made to an unfamiliar, newly-registered domain (randomdomain123.xyz), and a burst of over 2,000 SYN packets confirms port-scanning activity against 192.168.56.101. Neither the domain nor the IP is currently flagged as malicious by VirusTotal, but the domain's recent registration date is a mild indicator worth noting independent of that clean result.